

Course Description: Academic Year 2026-27

Course Title and Code: Fundamentals of Electrical and Electronics Engineering EE1118	
Teaching Scheme and no. of hours per week	L-T-P: 3-0-2 (5 hours)
Credits	4
Programme(s) and Batch	BTech
Semester (as per curriculum structure)	I Sem
NCrF Level (4.5/5.0/5.5/6.0)	5.5
Pre-requisite course (Course Code and Title)	Nil
Desired prior knowledge/skills	Basic Math
Weightage	Theory 60%; Practical 40%
The occupational path it may lead to, if any	-
Course Objective: This course aims to provide conceptual knowledge about electrical circuits and electronic devices and their real-life applications.	
Course Outcomes: On successful completion of this course, the students should be able to:	
EE1118.1 Understand the behavior of active and passive components.	
EE1118.2 Examine the transient and steady state response of DC circuits	
EE1118.3 Use diodes for rectification and voltage regulation.	
EE1118.4 Design switching circuits using BJT.	
EE1118.5 Use Operational Amplifiers in inverting and non-inverting configurations.	

Syllabus

Unit 1: Introduction to discrete components and abstractionns on which design of electronic circuits are based. Circuit Analysis Techniques- KVL, KCL, Source Transformations, Network theorems.

Unit 2: Sinusoidal steady state analysis of circuits containing resistors, capacitors, and inductors. Differential equations describe time and frequency behavior of circuits containing storge elements.

Unit 3: Semiconductor, Diode, PN junctions, Diode equation, models and applications i.e. half rectifier, full wave rectifier and Zener diode.

Unit 4: BJT construction and biasing, working, BJT as a switch, Amplifiers- Transconductance Magnitude and Phase response.

Unit 5: Operational amplifier and its applications as inverting and noninverting amplifiers, Op-amp as Buffer and Comparator.

Text/Reference Books

1. Engineering Circuit Analysis by WH Hayat, JE Kemmerly, SM Durbin, Tata Mc Graw Hill Education Private Limited.
2. Microelectronics Circuits by Adel S Sedra and Kenneth C Smith, Oxford Press.

Online Course:

<https://www.youtube.com/playlist?list=PLTFMW-rP7fuqMiK174et4SKbbMKd4VjdU>

Evaluation Scheme

Sr. No	Specifications	Weightage	Marks
1	Class Participation	05	05
2	Assignment	Nil	Nil
3	Quiz	15	15

4	Mid-Term Examination (Theory)	20	20
5	End-Term Examination (Theory)	30	30
6	Report	Nil	Nil
7	Project	10	10
8	Lab Evaluation	20	20
9	Course Portfolio	Nil	Nil
10	Presentation	Nil	Nil
11	Viva	Nil	Nil
	Total	100	100

Course Articulation Matrix: (Mapping of COs with POs)

Course Outcomes	Correlation with Program Outcomes															Correlation with Program-Specific Outcomes	
	PO 1	PO 2a	PO 2b	PO 2c	PO 3a	PO 3b	PO 3c	PO 4a	PO 4b	PO 4c	PO 5a	PO 5b	PO 6	PO 7a	PO 7b	PSO 1	PSO 2
CS1203.1					2												
CS1203.2								2									
CS1203.3																1	
CS1203.4	1				2						2					1	
CS1203.5	1																1

1- Low Correlation; 2- Moderate Correlation; 3- Substantial Correlation

List of Activities/Content with direct bearing on Employability/ Entrepreneurship/ Skill development

S. No.	Activity/Content	Skills	Tools/Methodology (AI Tools/Software/Presentation etc.)
1.	Familiarization with Laboratory Instruments: Oscilloscope, Function Generators, Digital Multimeter and DC Power supply	Instrument handling, measurement techniques, laboratory safety	Lab Equipments
2.	Source Transformation	Circuit analysis and simplification	Class Practice
3.	Experimental verification of electrical circuit problems using Kirchhoff's voltage laws and Kirchhoff's current laws.	Circuit construction, and analysis	Lab components
4.	Simulation and Experimental validation of electrical circuit problems using Kirchhoff's voltage laws and Kirchhoff's current laws.	Circuit simulation, result comparison, experimental validation	Microcap Simulator
5.	Experimental verification of electrical circuit problems using Thevenin Theorem.	Network analysis using simplified circuits	Lab components
6.	Electric circuit analysis using Theorems	Interpretation of electrical circuit behavior	Class Practice

7.	Simulation and Experimental validation of R-L and R-C circuits using Microcap	Transient response analysis, simulation, waveform interpretation	Microcap Simulator
8.	Transient analysis of R-L and R-C circuits	Transient response analysis, simulation, waveform interpretation	Class Practice
9.	Plot V-I characteristics of P-N junction diode using Microcap & calculate cut-in voltage, reverse Saturation current, and static & dynamic resistances.	Semiconductor device testing and parameter estimation	Microcap Simulator
10.	Design and analyze a half-wave and a full-wave rectifier.	Rectifier design, waveform analysis, power supply fundamentals	Lab components
11.	Diode Modelling and Analysis	Analyze diode models	Class practice
12.	PROJECT.	Problem-solving, teamwork, technical documentation, presentation skills	
13.	Plot the input and output characteristics of BJT in CE configurations.	Transistor characterization	Microcap Simulator
14.	Design an Op-Amp in inverting and non-inverting modes.	Analog circuit design, gain calculation and circuit implementation	Lab components

LESSON PLANNING (Theory)

Lecture	Unit/ Chapter/ Topic	Hours
Unit I		12
1	Introduction to discrete components and abstractions	
2	Source Transformations	
3	Circuit Analysis using KVL	
4	Circuit Analysis using KCL	
5	Network theorems: Thevenin Theorem	
6	Network theorems: Norton's Theorem	
7	Network theorems: Maximum Power Transfer Theorem	
Unit II		08
8	Analysis of R, L and C Elements under Sinusoidal Excitation	
9	Differential Equations of RL Circuits, Time Constant, Transient and Steady-State Response, Frequency Response	
10	Differential Equations of RC Circuits, Time Constant, Transient and Steady-State Response, Frequency Response	
Unit III		07
11	Semiconductor Fundamentals, PN Junction Diode	
12	Diode Equation and Characteristics	
13	Diode Models and Analysis	
14	Half-Wave and Full-Wave Rectifiers	

15	Zener diode	07
Unit IV		
16	BJT Construction, Working Principle and Biasing	
17	BJT Biasing	
18	BJT as a Switch	
19	BJT as an Amplifiers- Transconductance Magnitude and Phase response.	
Unit V		06
20	Introduction to Op-Amp, Ideal Characteristics	
21	Inverting and Non-Inverting Amplifiers	
22	Op-Amp as Buffer (Voltage Follower) and Comparator	

LESSON PLANNING (Practical)

Lab	Laboratory
1	Familiarization with Laboratory Instruments: Oscilloscope, Function Generators, Digital Multimeter and DC Power supply
2.	Experimental verification of electrical circuit problems using Kirchhoff's voltage laws and Kirchhoff's current laws.
3.	Simulation and Experimental validation of electrical circuit problems using Kirchhoff's voltage laws and Kirchhoff's current laws.
4.	Experimental verification of electrical circuit problems using Thevenin Theorem.
5.	Simulation and Experimental validation of R-L and R-C circuits using Microcap
6.	Plot V-I characteristics of P-N junction diode using Microcap & calculate cut-in voltage, reverse Saturation current, and static & dynamic resistances.
7.	Design and analyze a half-wave and a full-wave rectifier.
8.	PROJECT.
9.	Plot the input and output characteristics of BJT in CE configurations.
10.	Design an Op-Amp in inverting and non-inverting modes.